

CodexArena

Autonomous Adversarial Chaos & Self-Healing Engine for Microservices

Transforming AI from a passive autocomplete assistant into an active cyber-resilience engineer that stress-tests, breaks, synthesizes atomic row-level hot-patches, and certifies 100% resilience.

Red Agent

50-concurrent TOCTOU & invariant chaos attacker

Blue Agent

Autonomous telemetry analyzer & on-disk hot-patcher

Tamper-Proof Cert

Cryptographic SHA-256 resilience certification

Why Current CI/CD Pipelines Fail in Production

01

Single-Threaded Blindspot

Unit tests execute in sequential, isolated memory with mock databases. They never reproduce the chaotic multi-threaded conditions, lock contention, or race windows of live traffic.

02

Silent Invariant Violations

Critical bugs like TOCTOU race conditions, negative balance drains, and duplicate replay exploits slip past linters because individual lines of code look completely syntactically innocent.

03

Billion-Dollar Disasters

From multi-billion dollar outages like CrowdStrike to decentralized financial double-spends, production failures cost enterprises millions before engineers can even locate the root cause.

Autonomous Dual-Agent Digital Colosseum

Drop any Python microservice into targets/ folder. CodexArena autonomously pits attacker against healer.

Red Agent (Hostile Adversary)

- > 50-concurrent request bursts exploiting TOCTOU read/write window
- > Negative balance injections creating ghost treasury overdrafts
- > Self-transfer balance doubling invariant violations
- > High-velocity idempotency replay attacks bypassing trace IDs
- > Simulates real-world hostile assault with millisecond precision

Blue Agent (Autonomous Healer)

- > Intercepts crash telemetry, balance drift, and corrupted state dumps
- > IBM Granite 3.0 (ibm/granite-3-8b-instruct) performs Root Cause Analysis
- > Synthesizes precision atomic row-level SQL & idempotency cache
- > Applies hot-patches directly to target files on disk (zero restart)
- > Re-launches adversarial battle to mathematically prove 100% resilience

Orchestrated via Model Context Protocol (FastMCP)

IBM Bob is the central autonomous commander driving the arena over 5 custom stdio FastMCP tools.

5 FastMCP Tools Registered in .bob/mcp.json

launch_arena_battle

Commands Red Agent to unleash multi-threaded exploit burst

get_crash_telemetry

Extracts balance drift, corrupted account IDs, and stack traces

apply_hot_patch

Blue Agent writes atomic UPDATE locks & patches files on disk

verify_resilience

Executes immediate secondary assault to verify 100% immunity

export_resilience_certificate

Issues cryptographically signed JSON & printable PDF certificate

IBM Bob Verified Metrics

242.8k

Context tokens evaluated without drift

25 / 25

Multi-step execution tasks completed

31.64

Bobcoins consumed (high efficiency)

Task ID:

4d6a28e9ae59ca148f27c3d07811d1b3

(Stored in bob_sessions/ deliverable)

Cyber War-Room Cockpit & Cryptographic Certification

War-Room Cockpit Features

Real-Time Combat Feed

Server-Sent Events (SSE) stream live attack latencies, overdrifts, and patch actions at sub-millisecond speeds.

0-100% Resilience Gauge

Instant visual feedback as balance drift drops resilience to 0% and verified healing locks at 100% immunity.

IBM Granite 3.0 RCA Card

Direct telemetry diagnosis from ibm/granite-3-8b-instruct with 99.8% synthesis confidence.

SHA-256 Tamper-Proof Cert

Cryptographic digest sealing timestamp, hardened file inventory, and vulnerability audit trail.



The Future of Autonomous Cyber-Resilience

\$0

Zero-Day Financial Losses

Completely eliminates financial double-spending, ghost overdrafts, and negative balance exploits prior to production software merges.

< 3s

Mean Time To Remediation

Reduces incident response from hours of manual engineering debugging to autonomous telemetry analysis and on-disk verification in seconds.

Multi-Lang

Extensible Roadmap

Extending the autonomous digital colosseum from Python microservices to Go, Rust, and Solidity/EVM financial smart contracts.

"AI shouldn't just autocomplete code — it should actively defend it."

CodexArena x IBM Bob 2.0 • Team CodeMateX • Ready for Production CI/CD DevSecOps